

Skyflow for Snowflake - Deployment Guide

Complete step-by-step guide to deploy Skyflow tokenization and detokenization for Snowflake using AWS Lambda.

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Prerequisites

Required Tools

- **Node.js 18+** - [Download](#)
- **AWS CLI** - [Install Guide](#)

- **Note:** This guide uses AWS CLI for all deployment steps. If you prefer Infrastructure as Code (IaC), you can adapt these steps to Terraform, CloudFormation, or AWS CDK.
- **AWS Account** with permissions for Lambda, API Gateway, IAM, Secrets Manager
- **Snowflake Account** with ACCOUNTADMIN role
- **Skyflow Account** with vault access and authentication credentials (JWT or API Key)

Required Information

Before starting, gather these values:

From Skyflow:

- Vault URL (e.g., <https://your-cluster-id.vault.skyflowapis.com>)
- **Authentication credentials** (choose one):
 - **JWT (Service Account):** Client ID, private key, token URL, key ID (recommended)
 - **API Key:** Bearer token
- Vault IDs for each data type (NAME, ID, DOB, SSN)
- Table and column names for each data type

From AWS:

- AWS Account ID
- AWS Region (recommend: [us-east-1](#))

From Snowflake:

- Account identifier (e.g., [ABC12345.us-east-1](#))
- Username with ACCOUNTADMIN role
- Password
- Database and schema names
- Warehouse name



Deployment Steps

Step 1: Configure AWS CLI

```
# Configure AWS credentials
aws configure

# Enter:
# - AWS Access Key ID
# - AWS Secret Access Key
# - Default region (use us-east-1)
# - Default output format (json)

# Verify configuration
aws sts get-caller-identity
```

Step 2: Prepare Configuration Files

2.1 Create configuration file

Create `skyflow-config.json` with your Skyflow credentials. You can use either **JWT (Service Account)** or **API Key** authentication. The system will auto-detect which credentials are present (JWT is checked first).

Option A: JWT (Service Account) Authentication (recommended)

```
{
  "credentials": {
    "clientID": "YOUR_CLIENT_ID",
    "clientName": "YOUR_SERVICE_ACCOUNT_NAME",
    "tokenURI": "https://manage.skyflowapis.com/v1/auth/sa/oauth/token",
    "keyID": "YOUR_KEY_ID",
    "privateKey": "-----BEGIN PRIVATE KEY-----\nYOUR_PRIVATE_KEY_HERE\n---
--END PRIVATE KEY-----\n",
    "keyAlgorithm": "KEY_ALG_RSA_2048"
  },
  "vaults": {
    "vaultUrl": "https://YOUR_CLUSTER_ID.vault.skyflowapis.com",
    "definitions": [
      {
        "vaultId": "YOUR_VAULT_ID_FOR_NAMES",
        "table": "persons",
        "column": "name",
        "dataType": "NAME"
      },
      {
        "vaultId": "YOUR_VAULT_ID_FOR_IDS",
        "table": "persons",
        "column": "person_id",
        "dataType": "ID"
      },
      {
        "vaultId": "YOUR_VAULT_ID_FOR_DOBS",
        "table": "persons",
        "column": "date_of_birth",
        "dataType": "DOB"
      },
      {
        "vaultId": "YOUR_VAULT_ID_FOR_SSNS",
        "table": "persons",
        "column": "ssn",
        "dataType": "SSN"
      }
    ]
  },
  "tokenizeBatchSize": 5,
  "tokenizeMaxConcurrency": 400,
  "detokenizeBatchSize": 300,
}
```

```

    "detokenizeMaxConcurrency": 400,
    "logLevel": "INFO"
  }

```

Option B: API Key Authentication

```

{
  "credentials": {
    "apiKey": "YOUR_SKYFLOW_API_KEY_HERE"
  },
  "vaults": {
    "vaultUrl": "https://YOUR_CLUSTER_ID.vault.skyflowapis.com",
    "definitions": [
      {
        "vaultId": "YOUR_VAULT_ID_FOR_NAMES",
        "table": "persons",
        "column": "name",
        "dataType": "NAME"
      },
      {
        "vaultId": "YOUR_VAULT_ID_FOR_IDS",
        "table": "persons",
        "column": "person_id",
        "dataType": "ID"
      },
      {
        "vaultId": "YOUR_VAULT_ID_FOR_DOBS",
        "table": "persons",
        "column": "date_of_birth",
        "dataType": "DOB"
      },
      {
        "vaultId": "YOUR_VAULT_ID_FOR_SSNS",
        "table": "persons",
        "column": "ssn",
        "dataType": "SSN"
      }
    ]
  },
  "tokenizeBatchSize": 5,
  "tokenizeMaxConcurrency": 400,
  "detokenizeBatchSize": 300,
  "detokenizeMaxConcurrency": 400,
  "logLevel": "INFO"
}

```

Configuration Notes:

- Replace all **YOUR_*** placeholder values with your actual Skyflow configuration
- The **vaultUrl** should include your cluster ID (e.g., <https://abc123.vault.skyflowapis.com>)

- Batch sizes and concurrency can be tuned based on your workload (see [Configuration Options](#))
-

Deployment Mode Selection

This guide covers **Secrets Manager deployment** (recommended for production). There are two deployment modes available:

Option 1: Secrets Manager (This Guide) Recommended

Pros:

- Configuration updates without Lambda redeployment
- Centralized secrets management with audit logging
- Automatic encryption at rest
- Supports credential rotation with 5-minute cache refresh

Cons:

- Additional AWS service dependency
- Minimal cost (~\$0.40/month per secret)
- Requires IAM permissions for Secrets Manager

Use when: You need configuration flexibility, credential rotation, or are deploying to production.

Option 2: File-Based Configuration

In this mode, `skyflow-config.json` is bundled directly into the Lambda deployment package.

Pros:

- Simpler setup (no Secrets Manager)
- No additional AWS costs
- Faster cold starts (no Secrets Manager API call)

Cons:

- Configuration changes require Lambda redeployment
- Credentials stored in Lambda package (less secure)
- No centralized secrets management

Use when: You're doing development/testing or have a simple deployment with infrequent config changes.

To deploy with file-based config:

```
# Place your config in the lambda directory
cp skyflow-config.json lambda/

# Run the deployment script with 'config' argument
cd lambda
./deploy.sh config
```

This guide continues with **Secrets Manager deployment**. If you chose file-based, skip to Step 4 after running the deploy script.

Step 3: Create AWS Secrets Manager Secret

```
# Create the secret in us-east-1
aws secretsmanager create-secret \
  --name skyflow-tokenization-config \
  --description "Skyflow tokenization configuration" \
  --secret-string file://skyflow-config.json \
  --region us-east-1

# Verify secret was created
aws secretsmanager describe-secret \
  --secret-id skyflow-tokenization-config \
  --region us-east-1
```

Expected output:

```
{
  "ARN": "arn:aws:secretsmanager:us-east-1:YOUR_ACCOUNT:secret:skyflow-
tokenization-config-XXXXXX",
  "Name": "skyflow-tokenization-config",
  "Description": "Skyflow tokenization configuration",
  ...
}
```

Step 4: Create Lambda IAM Role

4.1 Create trust policy

Create `lambda-trust-policy.json`:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "Service": "lambda.amazonaws.com"
      },
      "Action": "sts:AssumeRole"
    }
  ]
}
```

4.2 Create the role

```
# Replace YOUR_ACCOUNT_ID with your AWS account ID
export AWS_ACCOUNT_ID=$(aws sts get-caller-identity --query Account --
output text)
export AWS_REGION=us-east-1

# Create IAM role
aws iam create-role \
  --role-name skyflow-tokenization-lambda-role \
  --assume-role-policy-document file://lambda-trust-policy.json \
  --description "Execution role for Skyflow tokenization Lambda"

# Attach basic Lambda execution policy
aws iam attach-role-policy \
  --role-name skyflow-tokenization-lambda-role \
  --policy-arn arn:aws:iam::aws:policy/service-
role/AWSLambdaBasicExecutionRole
```

4.3 Add Secrets Manager permissions

Create `secrets-manager-policy.json`:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": [
        "secretsmanager:GetSecretValue"
      ],
      "Resource": "arn:aws:secretsmanager:us-east-1:*:secret:skyflow-
tokenization-config-*"
    }
  ]
}
```

```
# Attach Secrets Manager policy
aws iam put-role-policy \
  --role-name skyflow-tokenization-lambda-role \
  --policy-name SecretsManagerReadPolicy \
  --policy-document file://secrets-manager-policy.json

# Wait for role to propagate
sleep 10
```

Step 5: Build Lambda Deployment Package

5.1 Install dependencies

```
# Make sure you're in the directory with the Lambda code
# Directory should contain: config.js, skyflow-client.js, handler.js,
package.json

npm install --production
```

5.2 Create deployment ZIP

```
# Create deployment package
zip -r function.zip config.js skyflow-client.js handler.js package.json
node_modules/

# Verify package size (should be ~5-10 MB)
ls -lh function.zip
```

Step 6: Create Lambda Function

```
# Create Lambda function
aws lambda create-function \
  --function-name skyflow-tokenization \
  --runtime nodejs20.x \
  --role arn:aws:iam::${AWS_ACCOUNT_ID}:role/skyflow-tokenization-
lambda-role \
  --handler handler.handler \
  --zip-file fileb://function.zip \
  --timeout 60 \
  --memory-size 256 \
  --description "Skyflow tokenization and detokenization for Snowflake"
\
  --environment Variables="{SECRETS_MANAGER_SECRET_NAME=skyflow-
tokenization-config}" \
  --region ${AWS_REGION}

# Verify Lambda was created
aws lambda get-function \
  --function-name skyflow-tokenization \
  --region ${AWS_REGION} \
  --query 'Configuration.[FunctionName,Runtime,Handler,State]'
```

Expected output:


```
[
  "skyflow-tokenization",
  "nodejs20.x",
  "handler.handler",
  "Active"
]
```

Step 7: Create API Gateway

7.1 Create REST API

```
# Create API Gateway
API_ID=$(aws apigateway create-rest-api \
  --name skyflow-tokenization-api \
  --description "API for Skyflow tokenization and detokenization" \
  --endpoint-configuration types=REGIONAL \
  --region ${AWS_REGION} \
  --query 'id' \
  --output text)

echo "API ID: ${API_ID}"

# Get root resource ID
ROOT_RESOURCE_ID=$(aws apigateway get-resources \
  --rest-api-id ${API_ID} \
  --region ${AWS_REGION} \
  --query 'items[?path==`/`].id' \
  --output text)

echo "Root Resource ID: ${ROOT_RESOURCE_ID}"
```

7.2 Create /tokenize resource

```
# Create /tokenize resource
TOKENIZE_RESOURCE_ID=$(aws apigateway create-resource \
  --rest-api-id ${API_ID} \
  --parent-id ${ROOT_RESOURCE_ID} \
  --path-part tokenize \
  --region ${AWS_REGION} \
  --query 'id' \
  --output text)

echo "Tokenize Resource ID: ${TOKENIZE_RESOURCE_ID}"
```

7.3 Create /tokenize/{datatype} resources

```
# Create sub-resources for each data type
for DATA_TYPE in name id dob ssn; do
    echo "Creating /tokenize/${DATA_TYPE}"

    RESOURCE_ID=$(aws apigateway create-resource \
        --rest-api-id ${API_ID} \
        --parent-id ${TOKENIZE_RESOURCE_ID} \
        --path-part ${DATA_TYPE} \
        --region ${AWS_REGION} \
        --query 'id' \
        --output text)

    # Create POST method
    aws apigateway put-method \
        --rest-api-id ${API_ID} \
        --resource-id ${RESOURCE_ID} \
        --http-method POST \
        --authorization-type NONE \
        --region ${AWS_REGION}

    # Set up Lambda integration
    aws apigateway put-integration \
        --rest-api-id ${API_ID} \
        --resource-id ${RESOURCE_ID} \
        --http-method POST \
        --type AWS_PROXY \
        --integration-http-method POST \
        --uri "arn:aws:apigateway:${AWS_REGION}:lambda:path/2015-03-
31/functions/arn:aws:lambda:${AWS_REGION}:${AWS_ACCOUNT_ID}:function:skyfl
ow-tokenization/invocations" \
        --region ${AWS_REGION}
done
```

7.4 Create /detokenize resource

```
# Create /detokenize resource
DETOKENIZE_RESOURCE_ID=$(aws apigateway create-resource \
    --rest-api-id ${API_ID} \
    --parent-id ${ROOT_RESOURCE_ID} \
    --path-part detokenize \
    --region ${AWS_REGION} \
    --query 'id' \
    --output text)

echo "Detokenize Resource ID: ${DETOKENIZE_RESOURCE_ID}"
```

7.5 Create /detokenize/{datatype} resources

```
# Create sub-resources for each data type
for DATA_TYPE in name id dob ssn; do
    echo "Creating /detokenize/${DATA_TYPE}"

    RESOURCE_ID=$(aws apigateway create-resource \
        --rest-api-id ${API_ID} \
        --parent-id ${DETOKENIZE_RESOURCE_ID} \
        --path-part ${DATA_TYPE} \
        --region ${AWS_REGION} \
        --query 'id' \
        --output text)

    # Create POST method
    aws apigateway put-method \
        --rest-api-id ${API_ID} \
        --resource-id ${RESOURCE_ID} \
        --http-method POST \
        --authorization-type NONE \
        --region ${AWS_REGION}

    # Set up Lambda integration
    aws apigateway put-integration \
        --rest-api-id ${API_ID} \
        --resource-id ${RESOURCE_ID} \
        --http-method POST \
        --type AWS_PROXY \
        --integration-http-method POST \
        --uri "arn:aws:apigateway:${AWS_REGION}:lambda:path/2015-03-
31/functions/arn:aws:lambda:${AWS_REGION}:${AWS_ACCOUNT_ID}:function:skyfl
ow-tokenization/invocations" \
        --region ${AWS_REGION}
done
```

7.6 Grant API Gateway permission to invoke Lambda

```
# Add Lambda permission for API Gateway
aws lambda add-permission \
    --function-name skyflow-tokenization \
    --statement-id apigateway-invoke \
    --action lambda:InvokeFunction \
    --principal apigateway.amazonaws.com \
    --source-arn "arn:aws:execute-
api:${AWS_REGION}:${AWS_ACCOUNT_ID}:${API_ID}/*" \
    --region ${AWS_REGION}
```

7.7 Deploy API

```
# Deploy API to prod stage
aws apigateway create-deployment \
```

```

--rest-api-id ${API_ID} \
--stage-name prod \
--description "Production deployment" \
--region ${AWS_REGION}

# Get API URL
API_URL="https://${API_ID}.execute-api.${AWS_REGION}.amazonaws.com/prod"
echo "API URL: ${API_URL}"

```

Step 8: Create Snowflake IAM Role

8.1 Create trust policy (placeholder)

Create `snowflake-trust-policy.json`:

```

{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::YOUR_ACCOUNT_ID:root"
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "sts:ExternalId": "PLACEHOLDER_WILL_BE_UPDATED"
        }
      }
    }
  ]
}

```

Replace `YOUR_ACCOUNT_ID` with your AWS account ID, then:

```

# Create Snowflake IAM role
aws iam create-role \
  --role-name SnowflakeAPIRole \
  --assume-role-policy-document file://snowflake-trust-policy.json \
  --description "IAM role for Snowflake API integration"

```

8.2 Attach API Gateway invoke policy

Create `api-invoke-policy.json`:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Action": "execute-api:Invoke",
      "Resource": "arn:aws:execute-api:us-east-1:YOUR_ACCOUNT_ID:YOUR_API_ID/*"
    }
  ]
}
```

Replace `YOUR_ACCOUNT_ID` and `YOUR_API_ID`, then:

```
# Attach policy to role
aws iam put-role-policy \
  --role-name SnowflakeAPIRole \
  --policy-name APIGatewayInvokePolicy \
  --policy-document file://api-invoke-policy.json

# Get role ARN (save this for Snowflake)
SNOWFLAKE_ROLE_ARN=$(aws iam get-role \
  --role-name SnowflakeAPIRole \
  --query 'Role.Arn' \
  --output text)

echo "Snowflake Role ARN: ${SNOWFLAKE_ROLE_ARN}"
```

Step 9: Configure Snowflake

9.1 Create API Integration

Connect to Snowflake and run:

```
USE ROLE ACCOUNTADMIN;

CREATE OR REPLACE API INTEGRATION skyflow_api_integration
  API_PROVIDER = aws_api_gateway
  API_AWS_ROLE_ARN =
  'arn:aws:iam::YOUR_ACCOUNT_ID:role/SnowflakeAPIRole'
  ENABLED = TRUE
  API_ALLOWED_PREFIXES = ('https://YOUR_API_ID.execute-api.us-east-1.amazonaws.com/');

-- Get trust policy values
DESC API INTEGRATION skyflow_api_integration;
```

9.2 Update AWS IAM Trust Policy

From the Snowflake output, copy:

- `API_AWS_IAM_USER_ARN`
- `API_AWS_EXTERNAL_ID`

Update `snowflake-trust-policy.json`:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Effect": "Allow",
      "Principal": {
        "AWS": "arn:aws:iam::123456789012:user/abc123-s"
      },
      "Action": "sts:AssumeRole",
      "Condition": {
        "StringEquals": {
          "sts:ExternalId": "ABC12345_SFCRole=1_xyz789"
        }
      }
    }
  ]
}
```

Update the trust policy in AWS:

```
aws iam update-assume-role-policy \
  --role-name SnowflakeAPIRole \
  --policy-document file://snowflake-trust-policy.json
```

9.3 Create External Functions

In Snowflake, run:

```
USE ROLE ACCOUNTADMIN;
USE DATABASE YOUR_DATABASE;
USE SCHEMA YOUR_SCHEMA;

-- Tokenization functions
CREATE OR REPLACE EXTERNAL FUNCTION TOK_NAME(plaintext VARCHAR)
  RETURNS VARCHAR
  API_INTEGRATION = skyflow_api_integration
  AS 'https://YOUR_API_ID.execute-api.us-east-
1.amazonaws.com/prod/tokenize/name';
```

```
CREATE OR REPLACE EXTERNAL FUNCTION TOK_ID(plaintext VARCHAR)
  RETURNS VARCHAR
  API_INTEGRATION = skyflow_api_integration
  AS 'https://YOUR_API_ID.execute-api.us-east-
1.amazonaws.com/prod/tokenize/id';

CREATE OR REPLACE EXTERNAL FUNCTION TOK_DOB(plaintext VARCHAR)
  RETURNS VARCHAR
  API_INTEGRATION = skyflow_api_integration
  AS 'https://YOUR_API_ID.execute-api.us-east-
1.amazonaws.com/prod/tokenize/dob';

CREATE OR REPLACE EXTERNAL FUNCTION TOK_SSN(plaintext VARCHAR)
  RETURNS VARCHAR
  API_INTEGRATION = skyflow_api_integration
  AS 'https://YOUR_API_ID.execute-api.us-east-
1.amazonaws.com/prod/tokenize/ssn';

-- Detokenization functions
CREATE OR REPLACE EXTERNAL FUNCTION DETOK_NAME(token VARCHAR)
  RETURNS VARCHAR
  API_INTEGRATION = skyflow_api_integration
  AS 'https://YOUR_API_ID.execute-api.us-east-
1.amazonaws.com/prod/detokenize/name';

CREATE OR REPLACE EXTERNAL FUNCTION DETOK_ID(token VARCHAR)
  RETURNS VARCHAR
  API_INTEGRATION = skyflow_api_integration
  AS 'https://YOUR_API_ID.execute-api.us-east-
1.amazonaws.com/prod/detokenize/id';

CREATE OR REPLACE EXTERNAL FUNCTION DETOK_DOB(token VARCHAR)
  RETURNS VARCHAR
  API_INTEGRATION = skyflow_api_integration
  AS 'https://YOUR_API_ID.execute-api.us-east-
1.amazonaws.com/prod/detokenize/dob';

CREATE OR REPLACE EXTERNAL FUNCTION DETOK_SSN(token VARCHAR)
  RETURNS VARCHAR
  API_INTEGRATION = skyflow_api_integration
  AS 'https://YOUR_API_ID.execute-api.us-east-
1.amazonaws.com/prod/detokenize/ssn';

-- Verify functions were created
SHOW FUNCTIONS LIKE 'TOK_%';
SHOW FUNCTIONS LIKE 'DETOK_%';
```

Testing

Test Lambda Directly

```
# Test tokenization
aws lambda invoke \
  --function-name skyflow-tokenization \
  --payload '{"path":"/tokenize/name","body":{"data":[[0,"John Doe"]]]}}' \
  --region us-east-1 \
  response.json

cat response.json
# Expected: {"statusCode":200,"body":{"data":[[0,"tok_..."]]]}}
```

Test in Snowflake

```
-- Test tokenization
SELECT TOK_NAME('John Doe') AS token;
-- Should return: tok_abc123...

-- Test detokenization
SELECT DETOK_NAME(TOK_NAME('John Doe')) AS original;
-- Should return: John Doe

-- Test all data types
SELECT
  TOK_NAME('Jane Smith') as name_token,
  TOK_ID('12345') as id_token,
  TOK_DOB('1990-01-01') as dob_token,
  TOK_SSN('123-45-6789') as ssn_token;

-- Test batch processing
SELECT
  customer_id,
  TOK_NAME(name) as name_token,
  TOK_SSN(ssn) as ssn_token
FROM your_table
LIMIT 100;
```

Check CloudWatch Logs

```
# View Lambda logs
aws logs tail /aws/lambda/skyflow-tokenization \
  --region us-east-1 \
  --follow

# Look for:
# - "Loading configuration..."
# - "Loading from AWS Secrets Manager: skyflow-tokenization-config"
# - "Initialized singleton Secrets Manager client" (first invocation only)
# - "Configuration loaded from Secrets Manager"
```



```
# - "Credentials type: Service Account (JWT)" or "Credentials type: API Key"
# - "Configuration validated successfully"
# - "Config cache expired, reloading from Secrets Manager" (after 5 minutes)
```

Performance Features

This implementation includes several optimizations for improved performance and reliability:

AWS Credential Management

- **Singleton Secrets Manager client:** Creates a single AWS Secrets Manager client per Lambda container
- Leverages AWS SDK's built-in credential refresh mechanisms
- Prevents race conditions during AWS IAM credential rotation
- Reduces API calls and improves cold start performance

Configuration Caching with TTL

- Configuration is cached for 5 minutes within each Lambda container
- Allows Skyflow credential rotation without Lambda restart
- Automatic reload after cache expiry
- **Promise-based locking** prevents multiple concurrent config loads during cache expiry
- Balances freshness with performance

Secrets Manager Resilience

- Exponential backoff with jitter for transient AWS errors (prevents thundering herd)
- Automatic retry for credential-related errors (`InvalidSignatureException`, `ExpiredTokenException`)
- Smart error classification (retryable vs non-retryable)
- 5xx server errors automatically retried (max 3 attempts)
- These retries ensure AWS infrastructure issues don't disrupt your operations

Skyflow SDK Integration

This implementation uses the **official Skyflow Node.js SDK v2.0.0**, which provides enterprise-grade features managed for you:

- **HTTP/2 connection management:** SDK handles connection multiplexing and header compression automatically
- **Built-in error handling:** Intelligent retry logic for Skyflow API errors
- **Optimized request batching:** Efficient payload formatting and parsing
- **Multi-vault support:** Seamlessly routes requests to correct vaults based on data type
- **Both authentication methods:** Supports JWT (Service Account) and API Key out of the box

These SDK features work transparently—you get the performance benefits without additional configuration.

Batch Processing

- Separate batch sizes for tokenization and detokenization
- Independent concurrency control for each operation
- Optimized for different workload patterns
- **Configurable batching**: Small batches (5) for low-latency tokenization, large batches (300) for high-throughput detokenization
- **Concurrency control**: Process multiple batches in parallel with configurable limits

Configuration Options

You can adjust performance settings in the Secrets Manager configuration. The Lambda function supports separate batch sizes and concurrency limits for tokenization and detokenization operations:

```
{
  "credentials": { ... },
  "vaults": { ... },
  "tokenizeBatchSize": 5,
  "tokenizeMaxConcurrency": 400,
  "detokenizeBatchSize": 300,
  "detokenizeMaxConcurrency": 400,
  "logLevel": "INFO"
}
```

Configuration Parameters:

- **tokenizeBatchSize**: Number of records to tokenize per batch (default: 5)
- **tokenizeMaxConcurrency**: Maximum concurrent tokenization batches (default: 400)
- **detokenizeBatchSize**: Number of records to detokenize per batch (default: 300)
- **detokenizeMaxConcurrency**: Maximum concurrent detokenization batches (default: 400)
- **logLevel**: Logging verbosity - **INFO**, **DEBUG**, or **ERROR** (default: INFO)

Tuning Guidelines:

- **Low latency tokenization**: Smaller batch size (5-10), higher concurrency (400-500)
- **High throughput detokenization**: Larger batch size (300-500), higher concurrency (300-400)
- **Rate limit sensitive**: Lower concurrency (100-200) for both operations
- **Balanced workload**: Default values work well for most use cases

Note: Configuration changes are picked up within 5 minutes due to the cache TTL. No Lambda redeployment is needed.

To update configuration:

```
# Update your local skyflow-config.json file, then:
aws secretsmanager update-secret \
  --secret-id skyflow-tokenization-config \
  --secret-string file://skyflow-config.json \
```

```
--region us-east-1

# Changes will be applied within 5 minutes (cache TTL)
```

Troubleshooting

Lambda Can't Read Secret

Error: `Missing vaultUrl in vaults configuration` or `Failed to load configuration from Secrets Manager`

Solution:

```
# Check Lambda environment variables
aws lambda get-function-configuration \
  --function-name skyflow-tokenization \
  --region us-east-1 \
  --query 'Environment.Variables'

# Should show:
# {
#   "SECRETS_MANAGER_SECRET_NAME": "skyflow-tokenization-config"
# }

# If missing, set it:
aws lambda update-function-configuration \
  --function-name skyflow-tokenization \
  --region us-east-1 \
  --environment Variables="{SECRETS_MANAGER_SECRET_NAME=skyflow-
tokenization-config}"
```

Permission Denied

Error: `AccessDeniedException` when calling `GetSecretValue`

Solution: Check Lambda role has Secrets Manager permissions:

```
aws iam get-role-policy \
  --role-name skyflow-tokenization-lambda-role \
  --policy-name SecretsManagerReadPolicy
```

HTTP 401 Unauthorized

Error: `HTTP 401: Unauthorized` or authentication failures

Solution: Verify your Skyflow credentials are valid in Secrets Manager:

```
# View secret structure (be careful – shows sensitive data)
aws secretsmanager get-secret-value \
  --secret-id skyflow-tokenization-config \
  --region us-east-1 \
  --query 'SecretString' \
  --output text | jq '.credentials'

# For JWT authentication, check:
# - clientID and privateKey are present
# - privateKey format is correct (PEM format with \n line breaks)
# - Service account has not been deleted or disabled in Skyflow

# For API Key authentication, check:
# - apiKey is present and valid
# - API key has not been revoked in Skyflow
```

Snowflake Functions Not Working

Error: Request failed for external function

Check these:

1. API Gateway URL is correct in function definition
2. Snowflake IAM role trust policy is updated
3. API Gateway deployment was created (**prod** stage)
4. Lambda has permission for API Gateway to invoke it

```
# Check Lambda permissions
aws lambda get-policy \
  --function-name skyflow-tokenization \
  --region us-east-1 \
  --query 'Policy' \
  --output text | jq .
```

Invalid Security Token

Error: The security token included in the request is invalid when accessing Secrets Manager

Cause: This error typically occurs during AWS IAM credential rotation. The implementation uses a singleton Secrets Manager client that works with AWS SDK's credential refresh to prevent this issue.

Solution:

1. Check Lambda execution role has correct permissions:

```
aws iam get-role \
  --role-name skyflow-tokenization-lambda-role \
```

```
--query 'Role.Arn'
```

2. Verify the role hasn't been deleted or policies detached
3. If error persists, check CloudWatch logs for more context:

```
aws logs tail /aws/lambda/skyflow-tokenization \  
  --region us-east-1 \  
  --follow
```

The singleton client pattern should prevent most occurrences of this error by allowing AWS SDK to manage credential lifecycle.

High Latency

Solution: Adjust performance settings in `skyflow-config.json`:

```
{  
  "credentials": { ... },  
  "vaults": { ... },  
  "tokenizeBatchSize": 10,  
  "tokenizeMaxConcurrency": 500,  
  "detokenizeBatchSize": 300,  
  "detokenizeMaxConcurrency": 400  
}
```

Then update the secret:

```
aws secretsmanager update-secret \  
  --secret-id skyflow-tokenization-config \  
  --secret-string file://skyflow-config.json \  
  --region us-east-1
```

Wait up to 5 minutes for the cache to refresh, then test again.

Updating the Lambda

To update the Lambda code:

```
# Make code changes, then rebuild  
npm install --production  
zip -r function.zip config.js skyflow-client.js handler.js package.json  
node_modules/
```

```
# Update Lambda
aws lambda update-function-code \
  --function-name skyflow-tokenization \
  --zip-file fileb://function.zip \
  --region us-east-1
```

Cleanup / Uninstall

To remove all resources:

```
# Delete Lambda function
aws lambda delete-function \
  --function-name skyflow-tokenization \
  --region us-east-1

# Delete API Gateway
aws apigateway delete-rest-api \
  --rest-api-id ${API_ID} \
  --region us-east-1

# Delete IAM roles
aws iam delete-role-policy \
  --role-name skyflow-tokenization-lambda-role \
  --policy-name SecretsManagerReadPolicy

aws iam detach-role-policy \
  --role-name skyflow-tokenization-lambda-role \
  --policy-arn arn:aws:iam::aws:policy/service-
role/AWSLambdaBasicExecutionRole

aws iam delete-role \
  --role-name skyflow-tokenization-lambda-role

aws iam delete-role-policy \
  --role-name SnowflakeAPIRole \
  --policy-name APIGatewayInvokePolicy

aws iam delete-role \
  --role-name SnowflakeAPIRole

# Delete secret
aws secretsmanager delete-secret \
  --secret-id skyflow-tokenization-config \
  --force-delete-without-recovery \
  --region us-east-1

# In Snowflake:
DROP FUNCTION IF EXISTS TOK_NAME(VARCHAR);
DROP FUNCTION IF EXISTS TOK_ID(VARCHAR);
DROP FUNCTION IF EXISTS TOK_DOB(VARCHAR);
```

```
DROP FUNCTION IF EXISTS TOK_SSN(VARCHAR);
DROP FUNCTION IF EXISTS DETOK_NAME(VARCHAR);
DROP FUNCTION IF EXISTS DETOK_ID(VARCHAR);
DROP FUNCTION IF EXISTS DETOK_DOB(VARCHAR);
DROP FUNCTION IF EXISTS DETOK_SSN(VARCHAR);
DROP API INTEGRATION IF EXISTS skyflow_api_integration;
```

Support

For issues or questions:

1. Check CloudWatch logs: `aws logs tail /aws/lambda/skyflow-tokenization --region us-east-1 --follow`
2. Review this troubleshooting section
3. Check Snowflake query history for error messages
4. Contact Skyflow support with CloudWatch log excerpts

Additional Resources

- **Skyflow API Documentation:** <https://docs.skyflow.com/>
- **AWS Lambda Documentation:** <https://docs.aws.amazon.com/lambda/>
- **AWS Secrets Manager:** <https://docs.aws.amazon.com/secretsmanager/>
- **Snowflake External Functions:** <https://docs.snowflake.com/en/sql-reference/external-functions-introduction>

Deployment complete! 🎉

Your Skyflow tokenization is now integrated with Snowflake using a Node.js Lambda function with performance optimizations including singleton client management, configuration caching, HTTP/2 support, buffer pooling, and adaptive retry logic.